IPC144 – Introduction to C Programming
Practice Exercises

Week 10 – Structure and Unions

Q1. Define a structure type `student` to represent a student. Include components for the following structure. Write functions to get student data elements from end user and display them on screen.

```
struct student
{
  int roll_no;
  char name[30];
  int phone_number;
};
```

Q2. Define a structure type `wallet` to represent a wallet. Include components for the following structure. Write functions to get wallet data elements from end user and display them on screen.

```
typedef struct wallet
{
    char name[200];
    unsigned long long int id;
} Wallet;

typedef struct advance_wallet
{
    char* name;
    unsigned long long int id;
} AdvanceWallet;
```

Q3. Define a structure type `auto_t` to represent an automobile. Include components for the make and model (strings), the odometer reading, the manufacture and purchase dates (use another user-defined type called date_t), and the gas tank (use a user-defined type tank_t with components for tank capacity and current fuel level, giving both in gallons). Write functions to get auto_t data elements from end user and display them on screen.

Q4. Define a structure type `account` to represent a bank account. Include components for the following structure. Write functions to get account data elements from end user and display them on screen.

```
typedef struct {
    int account_num;
    char account_type;
    char f_name[40];
    char l_name[40];
    float balance;
```

```
} account;
```

Q5. Define a structure type element_t to represent one element from the periodic table of elements. Components should include the atomic number (an integer); the name, chemical symbol, and class (strings); a numeric field for the atomic weight; and a seven-element array of integers for the number of electrons in each shell.

```
#define LEN_NAME 25 /* length of name string array in structure */
#define LEN_SYMBOL 5 /* length of symbol string array in structure */
#define LEN_CLAS 50 /* length of classification string array in structure */

typedef struct element_s {
     int anum; /* atomic number of element */
     char name[LEN_NAME], /* name of element */
     symbol[LEN_SYMBOL], /* chemical symbol of element */
     clas[LEN_CLAS]; /* classification of element */
     double weight; /* atomic weight of element */
     int electrons[7]; /* array representing # of electrons in each shell */
} element_t;
```

Q6. At a grocery store, certain categories of products sold have been established, and this information is to be computerized. Write a function to scan and store information in a structure variable whose data type is one you define—a type that includes a component that has multiple interpretations. Also, write an output function and a driver function to use in testing.

The data for each item consists of the item name (a string of less than 20 characters with no blanks), the unit cost in cents (an integer), and a character indicating the product category ('M' for meat, 'P' for produce, 'D' for dairy, 'C' for canned goods, and 'N' for nonfoods). The following additional data will depend on the product category.

Product Category	**Additional Data**
Meats	• character indicating meat type ('R' for red meat, 'P' for poultry,'F' for fi sh) • date of packaging • expiration date
Produce	• character 'F' for fruit or 'V' for vegetable • date received
Dairy	• expiration date
Canned goods	• expiration date (month and year only) • aisle number (an integer) • aisle side (letter 'A' or 'B')
Nonfoods	• character indicating category ('C' for cleaning product, 'P' for pharmacy, 'O' for other) • aisle number (an integer) • aisle side (letter 'A' or 'B')

A data line for canned corn would be

```
corn 89C 11 2000 12B
```

The corn costs 89 cents, expires in November of 2000, and is displayed in aisle 12B.